AyahVerse at HalluScoring-2026: Category-Aware Arabic Hallucination Detection with Multi-Layer Representations

Ibad-ur-Rehman Rashid
Department of Computer Science, Government Postgraduate College Mansehra, Pakistan
ibad.rashid.research@gmail.com

Abstract

We present our submission to HalluScoring 2026, an ArabicNLP shared task on hallucination detection and answer recovery in Arabic LLM outputs. For Track 1, our submitted baseline fine-tunes CAMeLBERT for binary hallucination detection, achieving 0.921 and 0.763 AUC-ROC on the development and test sets of Task 1.1, and 0.749 and 0.729 on Task 1.2, respectively. We further conduct post-submission ablations with intermediate-layer supervision and category embeddings to study their effect on generalization. The combined model achieves 0.929 AUC-ROC on Task 1.1 and 0.777 on Task 1.2 development sets. For Track 2, we separate hallucination detection from answer selection, using CAMeLBERT for detection and a decoder-based, QLoRA-tuned AceGPT-v2-8B-Chat for option selection. The system achieves combined scores of 0.784 on Task 2.1 and 0.829 on Task 2.2 on the validation sets, improving substantially over the corresponding zero-shot baselines.

1 Introduction

Arabic LLMs increasingly answer knowledge-intensive questions in Islamic scripture, history, science, and geography domains. Fluent but factually incorrect answers can pose a credibility risk, particularly in sensitive domains. HalluScoring 2026 (Alansari et al., 2026) evaluates this problem through two tracks: Track 1 detects whether a generated answer is hallucinated, while Track 2 additionally requires recovery of the verified answer from six candidate options.

We use CAMeLBERT (Inoue et al., 2021) for hallucination detection and AceGPT-v2-8B-Chat (Huang et al., 2024) for option selection. For Track 1, we fine-tune CAMeLBERT with intermediate-layer supervision and domain-category embeddings, evaluating their effects under in-domain and out-of-domain settings. For Track 2, we use the detector with options as input and pair it with a QLoRA-tuned AceGPT-v2-8B-Chat that selects candidate options through logit comparison. We participate in Leaderboard B (assisted detection), where the detector has access to the candidate options.

We made three contributions. First, we ran ablations over four training configurations: (1) a CAMeLBERT detector baseline, (2) the baseline with intermediate-layer supervision, (3) the baseline with category-aware representations, and (4) the baseline with both components for Arabic hallucination detection. Second, we analyze these components across Subtasks 1.1 and 1.2, which evaluate different generalization settings. Third, for Track 2 assisted detection, we retrain CAMeLBERT for hallucination detection and fine-tune AceGPT-v2-8B-Chat model with QLoRA (Dettmers et al., 2023) for option selection using logit-based scoring of candidate options. For Track 1 our system ranked second in Task 1.1, and third in Task 1.2. For Track 2 our system ranked third on both subtasks according to combined-score evaluation, while the leaderboard only mentions subtask 2.2 scores for our team. Code is available on GitHub[1]

2 Background

2.1 Task Setup

The HalluScoring 2026 shared task evaluates hallucination detection and factuality in Arabic LLM-generated answers (Alansari and Luqman, 2026). It comprises two tracks. Track 1 evaluates binary hallucination detection, while Track 2 combines hallucination detection with answer verification.

2.2 Data

Track 1: Hallucination Detection. Given an Arabic question and an LLM-generated answer,

[1]https://github.com/Ebad-urRehman/AyahVerse-at-HalluScoring-2026

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Subtask	Train	Dev	Test
1.1	4,705	1,300 (Seen)	1,030 (Seen)
1.2	4,705	200 (Unseen)	412 (Unseen)

Table 1: Track 1 dataset statistics.

Subtask	Domain	Train	Dev	Test
2.1	Islamic Knowledge	400	200	400
2.2	General Culture	1,600	800	1,600

Table 2: Track 2 dataset statistics.

the system predicts whether the answer is hallucinated. Subtask 1.1 evaluates generalization to unseen questions from seen models, whereas Subtask 1.2 evaluates generalization to answers generated by unseen models. The dataset sizes are summarized in Table 1.

Track 2: Detection-to-Truth. Track 2 combines hallucination detection with six-way answer selection. Each instance contains a question, generated answer, gold answer, and six candidate options. This track has two leaderboards: Blind Detection and Assisted Detection. We participate in Assisted Detection, where the detector has access to the candidate options.[2]

Subtask 2.1 focuses on Islamic knowledge, while Subtask 2.2 covers general culture and knowledge domains.

2.3 Related Work

Recent Arabic work has increasingly shifted toward fine-grained hallucination detection and analysis. General factuality research shows that fluent LLM outputs may still contain unsupported or incorrect information (Wang et al., 2024; Li et al., 2023). HalluScore (Alansari and Luqman, 2026) extends hallucination evaluation to Arabic knowledge-intensive and culturally grounded questions, while HalluTruthQA provides fine-grained factual annotations (Bouchekif et al., 2026) with 2,400 expert-curated Arabic QA instances containing verified answers, hallucination labels, erroneous spans, explanations, and hierarchical error types. HalluTruthQA-4K (Bekhouche et al., 2026) further extends this setting to 4,000 instances with verified answers and five plausible distractors, supporting both hallucination detection and factual verification. AlGhamdi et al. (2026) propose a knowledge-graph based diagnostic framework that analyzes hallucinations and question misalignment in Arabic Qur'anic machine reading comprehension through triple-level comparison of passage, question, and answer representations. These benchmarks motivate evaluating not only whether Arabic LLM outputs are incorrect, but also how factual errors can be detected and corrected.

Beyond detection, recovering the correct answer once a hallucination is identified requires efficient model adaptation. Parameter-efficient methods such as LoRA (Hu et al., 2021) and QLoRA (Dettmers et al., 2023) enable adaptation of large language models without full fine-tuning. HalluScoring 2026 (Alansari et al., 2026) brings these evaluation settings together through a shared task covering hallucination detection, generalization across models, and answer selection. Our contribution analyzes multi-layer representations and category embeddings for Arabic hallucination detection, and uses a decoder-based QLoRA-fine-tuned model for constrained answer recovery from the provided options. Together, these components address both the identification of hallucinated responses and the recovery of supported answers in a unified Arabic evaluation setting.

3 System Overview

3.1 Track 1: Category-Aware Multi-Layer Detection

1. Baseline configuration. We use CAMeLBERT (CAMeL-Lab/bert-base-arabic-camelbert-mix) with the default single-layer configuration. The submitted baseline uses the final-layer `[CLS]` representation h_{12}, which is passed to a binary classification head, with an auxiliary classification loss applied to the same layer. We then extend this baseline with intermediate-layer supervision and category embedding in post-submission ablation experiments.

2. Intermediate-layer supervision. We extract and concatenate the `[CLS]` representations h_4, h_8, and h_{12} from layers 4, 8, and 12 for classification, each layer supervised independently through its own auxiliary loss.

3. Category embedding. Each example is associated with a category indicated by its identifier. We extract the category prefix, e.g., `islamicHallucination_574` $\rightarrow$ `islamic`, and

[2] We report both 0.6/0.4 and 0.5/0.5 combined-score formulations in Table 5; the official leaderboard scores are marked separately.

Task	Split	Total	Valid	Default	Isl. / Hist. / Sci. / Geo.			
1.1	Train	4,705	1,600	3,105	400	400	400	400
1.1	Dev	1,300	776	524	176	200	200	200
1.1	Test	1,030	-	-	-	-	-	-
1.2	Train	4,705	1,600	3,105	400	400	400	400
1.2	Dev	200	0	200	0	0	0	0
1.2	Test	412	-	-	-	-	-	-

Table 3: Category Prefix Table. Valid examples contain a specific domain prefix, while Default examples have no recognizable prefix and fall back to index 0.

map it to a category index which is represented using a learned 32-dimensional embedding.

Category index assignment

`islamicHallucination`→ 1,
`historicHallucination`→ 2,
`ScienceHallucination`→ 3,
`geographyHallucination`→ 4

For identifiers containing only a numeric ID or an unrecognized prefix, category index 0 is used as the shared default category embedding. The category information is derived from the sample identifier rather than from the question or generated answer. The category embedding is concatenated with h_{12} when used alone, or with $[h_4; h_8; h_{12}]$ when combined with multi-layer supervision, with auxiliary losses applied at each selected layer.

4. Full Model: The full model combines multi-layer supervision and category embeddings, providing the final classifier with both multi-layer representations and explicit category information as shown in Figure 2.

3.2 Track 2: Decoder-based Answer Recovery

Track 2 combines hallucination detection with multiple-choice answer selection. We evaluate a zero-shot baseline and a separate-training pipeline that combines an encoder-based detector with a decoder-based option selector, as shown in Figure 1.

Zero-shot baseline. The zero-shot baseline combines pretrained CAMeLBERT detection with untuned AceGPT option scoring.

Separate training. Our system trains the detection and option-selection components separately. CAMeLBERT serves as the encoder-based detector, while the decoder-based AceGPT model is fine-tuned for option selection.

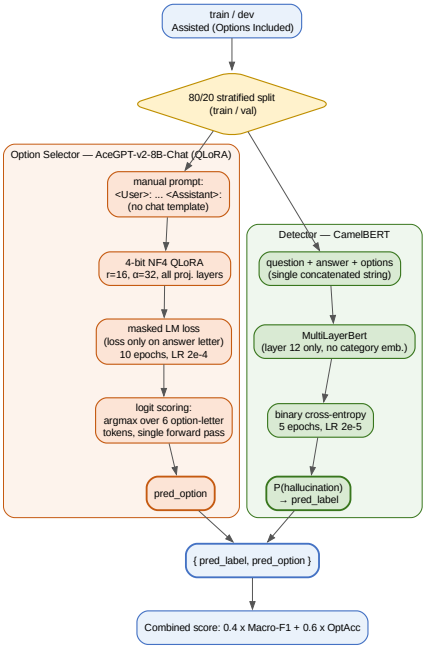


Figure 1: Track 2 system: detection and truth recovery.

Detection. We use the CAMeLBERT detector for the first stage. Because the assisted-detection setting exposes the candidate options to the system, the options are included in the detector input as:

السؤال : {question} الإجابة : {generated_answer} الخيارات: A: {option_A} B: {option_B} C: {option_C} D: {option_D} E: {option_E} F: {option_F}

Option selection. We fine-tune the decoder-based AceGPT-v2-8B-Chat with QLoRA for option selection. At inference time, candidate options are scored from their token logits using a single forward pass. We compare the logits assigned to the six candidate option tokens at the final prediction position and select the option with the highest score. This formulation avoids autoregressive generation and parsing failures while allowing the candidates to be scored efficiently in batches.

4 Experimental Setup

Track 1 training. We fine-tune CAMeLBERT for up to five epochs using a batch size of 16 and a learning rate of 2×10^{-5}. We use 10% warmup, a weight decay of 0.01, and FP16 training. Early stopping with a patience of three epochs is used. Inputs are formatted as `[CLS] question [SEP]`

`model_answer [SEP]` and truncated to a maximum length of 512 tokens.

The baseline training objective is

$$\mathcal{L} = \mathcal{L}_{\text{final}} + \alpha \mathcal{L}_{12}, \qquad \alpha = 0.3,$$

where $\mathcal{L}_{\text{final}}$ is the final classification loss produced by the final classifier, and $\mathcal{L}_{12}$ is the auxiliary classification loss produced by a separate classification head operating on h_{12}.

For intermediate-layer supervision, the objective is

$$\mathcal{L} = \mathcal{L}_{\text{final}} + \alpha \sum_{l \in \{4,8,12\}} \mathcal{L}_l, \qquad \alpha = 0.3,$$

where $\mathcal{L}_{\text{final}}$ is the cross-entropy loss from the classifier operating on the concatenated representation $[h_4; h_8; h_{12}]$, and $\mathcal{L}_l$ is the auxiliary loss produced by a separate classification head attached to layer l.

The auxiliary head is used only during training in both cases.

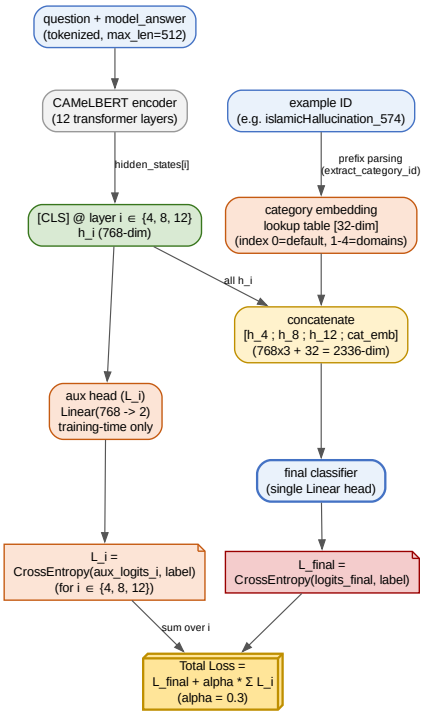


Figure 2: Full model: layer supervision + category embedding.

Track 2 detection. We use the CAMeLBERT pretrained (CAMeL-Lab/bert-base-arabic-camelbert-mix) checkpoint without task-specific fine-tuning as a baseline for assisted detection, where the candidate options are included in the input. For the fine-tuned CAMeLBERT, we use the same training hyperparameters as the Track 1 baseline.

Track 2 option selection. For option selection, we fine-tune AceGPT-v2-8B-Chat with QLoRA using 4-bit NF4 quantization, rank-16 adapters, and a learning rate of 2×10^{-4}.

As AceGPT does not provide a native chat template, we construct the instruction format manually. We also concatenate token IDs directly to avoid BPE boundary inconsistencies. Left padding and explicit position IDs are used to maintain RoPE alignment, while left-side truncation preserves the answer-label tokens required for option scoring.

For both Track 2 components, the training data is split 80/20 into training and internal validation sets, respectively, while the official development set remains unseen during training and model selection.

5 Results

Official versus post-submission results. Our official shared-task submission used the baseline CAMeLBERT configuration described in Section 3. The enhanced configurations were evaluated after submission on the development sets and were not submitted to the shared-task leaderboard.

The full model performs best overall on Task 1.1, while on Task 1.2, category embedding gives the highest AUC-ROC and the full model gives the highest Macro-F1, as shown in Table 4. For Track 2, the fine-tuned pipeline substantially outperforms the zero-shot baseline on both subtasks. Official test-set results are reported separately and marked in Table 5.

6 Analysis

Ablation Analysis The ablation results show that the contribution of each component depends on the evaluation setting. In Task 1.1, the individual components produce little change relative to the baseline, while their combination gives the strongest overall performance. In Task 1.2, category embedding provides the largest individual AUC-ROC improvement on the development set, from 0.749 to 0.782. Since all examples in this development set use the default category embedding, this improvement may reflect the large number of

Config	AUC-ROC	Macro-F1	AUC-PR
Task 1.1 (in-domain)			
Baseline (Dev. Set) †	0.921	0.860	0.935
Baseline (Test) †	0.763	0.640	0.918
+ layer supervision (Dev.)	0.921	0.854	0.940
+ category embedding (Dev.)	0.920	0.853	0.937
+ both (full model) (Dev.)	**0.929**	**0.867**	**0.948**
Task 1.2 (out-of-domain)			
Baseline (Dev. Set) †	0.749	0.632	0.844
Baseline (Test) †	0.729	0.613	0.908
+ layer supervision (Dev.)	0.768	0.658	0.857
+ category embedding (Dev.)	**0.782**	0.669	**0.868**
+ both (full model) (Dev.)	0.777	**0.703**	0.852

Table 4: Ablation results with full training data and dev-set evaluation. † denotes official shared-task leaderboard results.

Task	Approach	Comb. (0.6/0.4)	Comb. (0.5/0.5)	Det. AUC	Det. Acc.	Opt. Acc.	Macro-F1
2.1	Zero-shot (No Training) (Dev. Set)	0.578	0.611	0.815	0.841	0.445	0.778
	Fine-tuned Models (Dev. Set)	**0.784**	**0.791**	**0.960**	**0.855**	**0.755**	**0.828**
	Fine-tuned Models (Test Set)	**0.772**	**0.767**	**0.814**	**0.747**	**0.795**	**0.739**
2.2	Zero-shot (No Training) (Dev. Set)	0.649	0.650	0.745	0.655	0.646	0.654
	Fine-tuned Models (Dev. Set)	**0.829**	**0.828**	**0.917**	**0.831**	**0.835**	**0.821**
	Fine-tuned Models (Test Set)	**0.785**	**0.776**†	**0.848**	**0.766**	**0.819**	**0.734**

Table 5: Task 2 results for the validation and test sets under assisted detection. † denotes official shared-task leaderboard results.

training examples assigned to the default category (3,105) rather than domain-specific category information. The full model similarly achieves the highest Macro-F1 on this development set, from 0.632 to 0.703. Since the full model also incorporates the category embedding, this gain is subject to the same caveat.

In-Domain vs. Out-of-Domain Generalization. The difference between the two settings is more noticeable under out-of-domain evaluation. While the individual components have a minor effect on Task 1.1, both provide larger improvements on Task 1.2, particularly category embedding. However, the category-embedding improvement is measured on a development set where all examples use the shared default category embedding (see Table 3 for split-level coverage). Therefore, this improvement should not be interpreted as evidence of domain-specific category information improving generalization to unseen generators.

Decoder-Based Option Selection The fine-tuned option selector performs substantially better than zero-shot scoring, reaching 0.755 and 0.835 option accuracy on Tasks 2.1 and 2.2, respectively. This improvement is obtained with 4-bit NF4 QLoRA and direct comparison of candidate-token logits, avoiding free-form generation and output parsing.

Qualitative examples of model disagreements and representative errors are provided in Appendix A, while the study's limitations are discussed in Limitations section.

7 Conclusion

Separately, post-submission ablations show that the combined configuration achieves the strongest development-set performance on Task 1.1, reaching 0.929 AUC-ROC, while achieving 0.777 on Task 1.2. Our Track 2 system decouples detection from option selection, using a QLoRA-tuned decoder-based AceGPT model for option scoring, and achieves combined validation scores of 0.784 on Task 2.1 and 0.829 on Task 2.2. Future work includes Arabic-specific preprocessing, threshold calibration, and joint training of the detection and option-selection components.

Limitations

Our study has several limitations. First, the enhanced Track 1 configurations were evaluated only on development sets after submission, so their improvements have not been confirmed on a held-out test set. Second, we use a single run per configuration and do not report statistical significance. For Task 1.2, the category-embedding ablation is measured on a development set where all examples use the default category embedding, so its improvement may not reflect domain-specific category information, and more plausibly reflects the default embedding's own contribution as the most heavily-trained category vector (3,105 of 4,705 training examples). Evaluating the contribution of domain-specific category embeddings would require per-category ablation results on evaluation examples with identifiable category prefixes. Prefix coverage on the official Task 1.1/1.2 test sets could not be computed for this paper, as the underlying test IDs were not released to participants. Finally, Track 2 uses separate training for detection and option selection; joint training was not explored.

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A Error Analysis

The examples in Table 6 show three recurring patterns. First, intermediate-layer supervision can correct cases where the others miss a factually valid but concise answer, as in 51-M1. Second, category information can help with domain-specific cases, such as Sci-948 and Isl-92, where category embedding preserves the baseline's correct prediction, while layer supervision and the full model both fail on the same examples. Third, some fluent but incorrect responses remain difficult for all configurations, as illustrated by 51-M2. The examples also show that the enhanced configurations can correct specific factual errors, including incorrect scientific answers and erroneous citations, although the gains are not consistent across all cases.

Question | Arabic | Translation							
ID	**Generated Answer**	**True Label**	**Base**	**+Layer**	**+Cat.**	**Full**	**Note**
Question: ا نوع الأسلوب في جملة 'إياك والكذب *(What is the type of style in the sentence "Beware of lying"?)*							
51-M1	تحذير. ("Warning.")	Not hallucinated	✗	✓	✗	✗	Only layer supervision fixes it.
Question: ا نوع الأسلوب في جملة 'إياك والكذب *(What is the type of style in the sentence "Beware of lying"?)*							
51-M2	أسلوب إنشائي (نهي). (Rhetorical style (prohibition))	Hallucinated	✗	✗	✗	✗	The response is fluent but incorrect, and none of the configurations identifies the error.
Question: من الملقب بالملقب بشيخ تراجمة العصر العباسي *(Who is known as the "Sheikh of Translators" of the Abbasid era?)*							
79-M1	الملقب بشيخ تراجمة العصر العباسي هو حنين بن إسحاق (The person known as the "Sheikh of the Abbasid-era translators" is Hunayn bin Ishaq.)	Not hallucinated	✓	✓	✓	✓	All configurations got it right, question is very similar to answer.
Question: من الملقب بالملقب بشيخ تراجمة العصر العباسي *(Who is known as the "Sheikh of Translators" of the Abbasid era?)*							
79-M2	حنين بن إسحاق (Hunayn bin Ishaq)	Not hallucinated	✗	✗	✗	✗	All got wrong.
Question: A ماهي صفات الشخص الذي يحمل فصيلة الدم *(What are the characteristics of a person with blood type A?)*							
56-M2	فصيلة الدم A شائعة ولا ترتبط علمياً بصفات شخصية محددة؛... (Blood type A is common and is not scientifically linked to specific personality traits;...)	Not hallucinated	✗	✓	✗	✗	Only layer supervision fixes it.
Question: ما هو العنصر الأكثر وفرة في القشرة الأرضية *(What is the most abundant element in the Earth's crust?)*							
Sci 948	الأكسجين (Oxygen)	Not hallucinated	✓	✗	✓	✗	Baseline and Category both retain the correct prediction.
Question: ؟ من هم القوم الذين خرجوا من ديارهم وهم ألوف حذر الموت *(Who were the people who left their homes numbering...)*							
Isl 92	ي إسرائيل، كما ورد في القرآن الكريم في سورة البقرة، الآية 249 (Children of Israel, as stated in the Holy Quran in Surah Al-Baqarah, Verse 249...)	Hallucinated (Wrong Ayah citation and generation)	✓	✗	✓	✗	Baseline and Category both retain the correct prediction.

Table 6: Qualitative examples from Task 1.1 and 1.2 showing how the baseline and three enhanced configurations (layer supervision, category embedding, full model) disagree on individual predictions. ✓ = matches the true label, ✗ = does not.